## MC 451 Data Wrangling in R rubric

**50 points.** Due Friday, November 6, 11:59 PM.

**How this is scored.** Add the points from each row; the total is out of 50. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Submissions are unlimited before the deadline, so you can revise and submit again.

- Before you submit: restart R, run the script top to bottom, and see what breaks.

*Data Wrangling in R, 50 points*

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **1. Hand-code**<br>6 points | **6 points.** One variable from your codebook hand-coded for a 30-message sample drawn with `sample_messages()`, with the codes saved where the script can read them. | **5 points.** Hand-coded, but the sample was not drawn with `sample_messages()` or is not 30 messages. | **3 to 4 points.** Coded with a variable that is not in your codebook, or the codes are described but not saved. | **0 to 2 points.** No hand-coding. |
| **2. Load**<br>6 points | **6 points.** The chat and stream tables are loaded from the v2v package inside the script. | **5 points.** Both loaded, but from a file on your own computer, so the script fails anywhere else. | **3 to 4 points.** Only one table loaded. | **0 to 2 points.** Not loaded in the script. |
| **3. Convert the timestamps**<br>6 points | **6 points.** Epoch milliseconds converted to a real date-time, dividing by 1,000 first. | **5 points.** Converted, but a small error remains, such as times off by some hours because the time zone was not set. | **3 to 4 points.** Converted without dividing by 1,000, so the dates land thousands of years away. | **0 to 2 points.** Not converted. |
| **4. Message length**<br>6 points | **6 points.** A new column holding each message's length. | **5 points.** Length computed, but on the wrong field (for example, the username). | **3 to 4 points.** Length computed but not saved as a column. | **0 to 2 points.** Missing. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **5. Join**<br>6 points | **6 points.** Stream information joined onto the chat messages, with a comment naming the key you joined on. | **5 points.** Joined, but no comment names the key. | **3 to 4 points.** The join runs but matches nothing or multiplies the rows, and the script does not notice. | **0 to 2 points.** No join. |
| **6. Missing values**<br>6 points | **6 points.** Reports how many rows carry missing values and what you did with them. | **5 points.** Handled, but the count is not reported. | **3 to 4 points.** Rows with missing values dropped without saying how many. Dropping them silently is the error this step exists to catch. | **0 to 2 points.** Not handled. |
| **7. Export**<br>6 points | **6 points.** Saved as `twitch_analysis.RDS`. | **5 points.** Saved under another name, so the next two assignments cannot find it. | **3 to 4 points.** Saved as a CSV or another format that loses the column types. | **0 to 2 points.** Not exported. |
| **8. Every step is in the script**<br>6 points | **6 points.** Every change to the data happens in the script; nothing is done by hand in the console. | **5 points.** One step was done outside the script, but a comment says what and how. | **3 to 4 points.** Several steps were done outside the script. | **0 to 2 points.** The script does not rebuild the dataset. |
| **Clean-session run**<br>2 points | **2 points.** Runs start to finish after restarting R. | **1 point.** Runs after one small fix, such as a missing `library()` line. | **0 points.** Stops partway in a fresh session. | **0 points.** Runs only with objects already in your environment, or not at all. |